SMART FOOD SAFETY BOX

BE23GE311-DESIGN THINKING

REPORT

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ELECTRICAL AND ELECTRONICS ENGINEERING

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BONAFIDE CERTIFICATE

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ABSTRACT

Food spoilage poses significant health risks, particularly to individuals relying on leftover food near public places such as railway stations, hospitals, and temples. The absence of effective monitoring systems often results in the distribution of unsafe food, contributing to foodborne illnesses and unnecessary waste. The **Smart Food Safety Box** addresses this challenge through a real-time food monitoring and alert system designed to ensure the safe storage and distribution of surplus food

This innovative device integrates multiple sensors, including**MQ-4gas sensors** for detecting spoilage gases, **DHT11/DHT22** for humidity and temperature monitoring, and the **DS18B20 temperature sensor** for accurate thermal readings. Data from these sensors is processed and transmitted via the **ESP8266 Wi-Fi module** to a connected mobile application, which provides immediate alerts in case of spoilage. An **LCD display** shows real-time food condition metrics, while **servo motors** manage automated container operations to maintain hygiene and minimize manual contact.

Powered by a reliable **power supply module**, the Smart Food Safety Box can be deployed in various public settings to store and monitor leftover food safely. This solution promotes public health, reduces food waste, and supports sustainability goals such as **Zero Hunger** and **Good Health and Well-being** under the United Nations Sustainable Development Goals (SDGs).

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CHAPTER 1
INTRODUCTION TO DESIGN THINKING

Design Thinking is an innovative, human-cantered methodology that prioritizes understanding users' needs and experiences to develop effective solutions. This approach is defined by its iterative process, enabling teams to thoroughly explore challenges and generate creative solutions.

The Design Thinking process is generally comprised of five essential phases: Understand / Empathize, Observe, Define, Ideate, Prototype and Test.

UNDERSTAND / EMPATHIZE

This initial phase focuses on engaging with users through methods such as interviews and observations to gather insights into their thoughts, feelings, and experiences. Commonly used tools like Personas and Empathy Maps help visualize user needs and challenges effectively.

OBSERVE

This phase focuses on understanding users and their needs by watching and studying the min their natural environment. This involves gathering insights into their behaviors, needs, and pain points through various methods like observation, interviews, and surveys. The goal is to gain empathy for the user and uncover their true needs and desires, which are often not explicitly stated.

DEFINE

In the Define phase, the insights obtained during the Empathize phase are synthesized to create a clear and concise problem statement. The Hook Canvas can be utilized to identify user motivations and frame the problem from a user-centered perspective.

IDEATE

This phase promotes brainstorming and creative thinking to produce a broad range of ideas. Techniques such as mind mapping and the Figma Framework are employed to enhance collaborative ideation among team members.

PROTOTYPE

During the Prototype phase, teams develop tangible representations of their ideas, which can vary from low-fidelity sketches to high-fidelity models. Tools like Figma and Just in mind are used to visualize these concepts and assess their feasibility

TEST

The final phase focuses on collecting feedback on prototypes from actual users. Tools such as the AEIOU Tool assist in evaluating user interactions and refining solutions based on the insights gathered during testing. 2

By harnessing these phases and tools, Design Thinking fosters a collaborative atmosphere that promotes the creation of innovative solutions designed to address user needs effectively. This methodology proves to be particularly effective in overcoming complex challenges across various domains.

CHAPTER 2
PROBLEM STATEMENT

In many public areas such as railway stations, hospitals, and temples, leftover food is often collected and distributed to people in need. However, the lack of an effective monitoring system to assess the freshness and safety of this food leads to a high risk of spoilage, which can cause serious health issues. Furthermore, without proper detection methods, a significant portion of surplus food goes to waste instead of being safely reused. There is a critical need for a smart, real-time solution to monitor food quality, prevent health hazards, and promote safe food distribution to reduce hunger and minimize food wastage.

CHAPTER 3
UNDERSTAND PHASE

OVERVIEW

Many people near public places rely on leftover food, which is often stored without proper safety checks, leading to health risks from unnoticed spoilage. There is no reliable real-time monitoring system, and manual checks are inefficient. This increases the risk of contamination and waste. Addressing these issues is crucial for improving food safety and distribution.

TOOLS USED

1).User persona

INTERPRETATION

Food spoilage in public areas where leftover food is distributed poses serious health risks and leads to waste. People may unknowingly consume spoiled food, causing illness. There is no effective system to monitor food freshness in real time. Manual checks are unreliable and often miss early signs of spoilage. This highlights the urgent need for an automated solution to ensure food safety and reduce waste.

CHAPTER 4
OBSERVE PHASE

OVERVIEW

In the Observe phase, it was found that leftover food is stored without monitoring temperature, humidity, or spoilage signs. Volunteers rely on visual checks, which are often inaccurate. People consuming the food are unaware of its safety, risking health issues. These observations show the need for a system that ensures real-time food safety monitoring.

TOOLS USED

1.Empathy map

INTERPRETATION

In the Observe phase, it is evident that food storage in public areas lacks proper monitoring of spoilage factors like temperature and humidity. Surplus food is often checked manually, which is unreliable and may overlook early signs of spoilage. Volunteers face challenges ensuring food safety with limited tools. People consuming this food are at risk of health issues due to undetected spoilage. These observations highlight the need for a real-time monitoring system to ensure food safety and reduce waste.

CHAPTER 5
DEFINE PHASE

OVERVIEW

In the Define phase, the key problems identified are the lack of real-time monitoring and reliable methods to assess the safety of leftover food. People depending on such food are at risk due to unnoticed spoilage and poor storage practices. Manual checks are in effective and lead to both foodborne illnesses and unnecessary waste. There is a clear need for a smart, automated solution to ensure food safety and improve distribution.

TOOLS USED

Chart work- Gantt Template

INTERPRETATION

In the Define phase, insights from the problem and observation stages are synthesized to clearly state the core issue. It becomes evident that the absence of a reliable, real-time monitoring system is a major gap in ensuring food safety. Manual inspection methods are insufficient and contribute to both health risks and food waste. The needs of both consumers and distributors highlight the importance of a smart solution. This phase helps frame the problem as a need for an automated system to detect spoilage and ensure safe food distribution.

CHAPTER 6

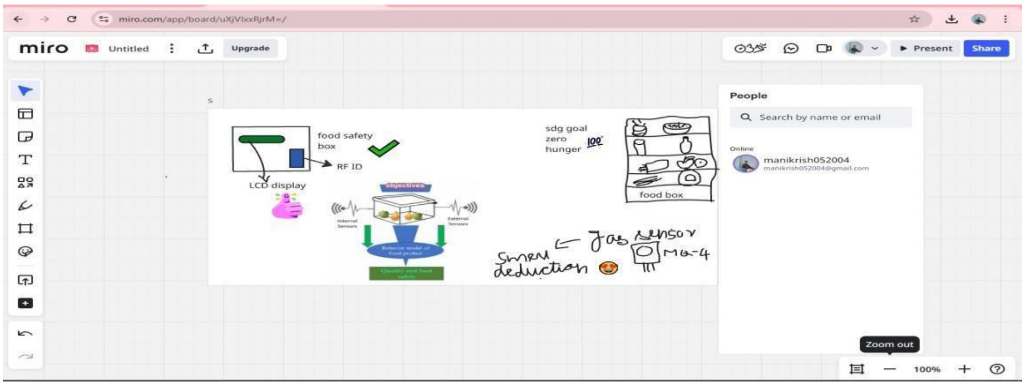
IDEATE PHASE

OVERVIEW

In the Ideate phase, the focus is on generating creative solutions to address the lack of food safety in leftover food distribution. Based on the defined problem, ideas are developed to monitor food conditions using technology like sensors and real-time alerts. Brainstorming leads to the concept of a smart system that ensures food freshness and reduces waste. This phase emphasizes innovation to improve both safety and efficiency in food distribution.

TOOLS USED

1).Miro



INTERPRETATION

In the Ideate phase, various solutions were explored to tackle the issue of food spoilage and unsafe distribution. The idea of integrating sensors to monitor temperature, humidity, and gas levels emerged as a practical and scalable approach. Real-time data transmission and alert systems were identified as key features to ensure food safety. The concept of automating food storage and monitoring gained traction as it minimizes human error. This phase clarifiedthat a smart, sensor-based system could effectively address the core challenges identified earlier.

CHAPTER 7
PROTOTYPE AND TEST PHASE

OVERVIEW

The prototype of the Smart Food Safety Box was developed to detect signs of food spoilage using temperature, humidity, and gas sensors. The device is intended to monitor stored food and send real-time alerts via Wi-Fi to a mobile app when food becomes unsafe for consumption. The system was built using commonly available microcontrollers and tested in simulated environments such as warm and humid containers to evaluate its performance.

TOOLS USED

1).Feed Back Sheet

INTERPRETATION

The prototype successfully detected temperature and gas changes in the food compartments. The sensors responded quickly to changes in food spoilage conditions. The mobile app received real-time notifications when unsafe conditions were detected. The servo motor mechanism worked well to secure the compartment. The results confirmed that the system could be scaled and deployed in real-world environments to help reduce food waste and ensure safe food distribution

CHAPTER 8
CONCLUSION

The Smart Food Safety Box project has demonstrated the feasibility of using affordable IoT-based technology to address real-world issues like food spoilage and hunger. The system integrates multiple sensors—such as temperature, humidity, and gas sensors—with a microcontroller (ESP8266) to monitor the quality of stored food in real time. When unsafe conditions are detected, the system instantly sends alerts to a mobile application, ensuring that action can be taken before the food becomes harmful for consumption.

This innovation is especially beneficial in public spaces where leftover food is often donated, such as railway stations, temples, hospitals, and shelter homes. By providing a smart storage solution, this project helps ensure that food meant for donation remains safe and hygienic, thus protecting the health of those in need.

The development and testing phases proved the effectiveness of the system under different simulated environments. The prototype successfully sent alerts and operated all hardware components reliably. Future scope includes adding features like AI-based spoilage prediction, GPS tracking for food boxes, solar-powered energy options, and a more robust mobile application.

In conclusion, the Smart Food Safety Box is not just a technical solution—it is a social innovation aimed at fostering food security, public health, and sustainability .With proper implementation and scale-up, it holds the potential to make a tangible impact in communities where access to safe food is a critical concern.

REFERENCE

How to connect ESP8266 with RFID

https://www.electronicwings.com/esp32/rfid-rc522-interfacing-with-esp32

Research about food spoilage detector

https://www.ijraset.com/research-paper/iot-basedfood-spoilage-detector

Food Waste and Distribution Problem

Food and Agriculture Organization (FAO) of the United Nations

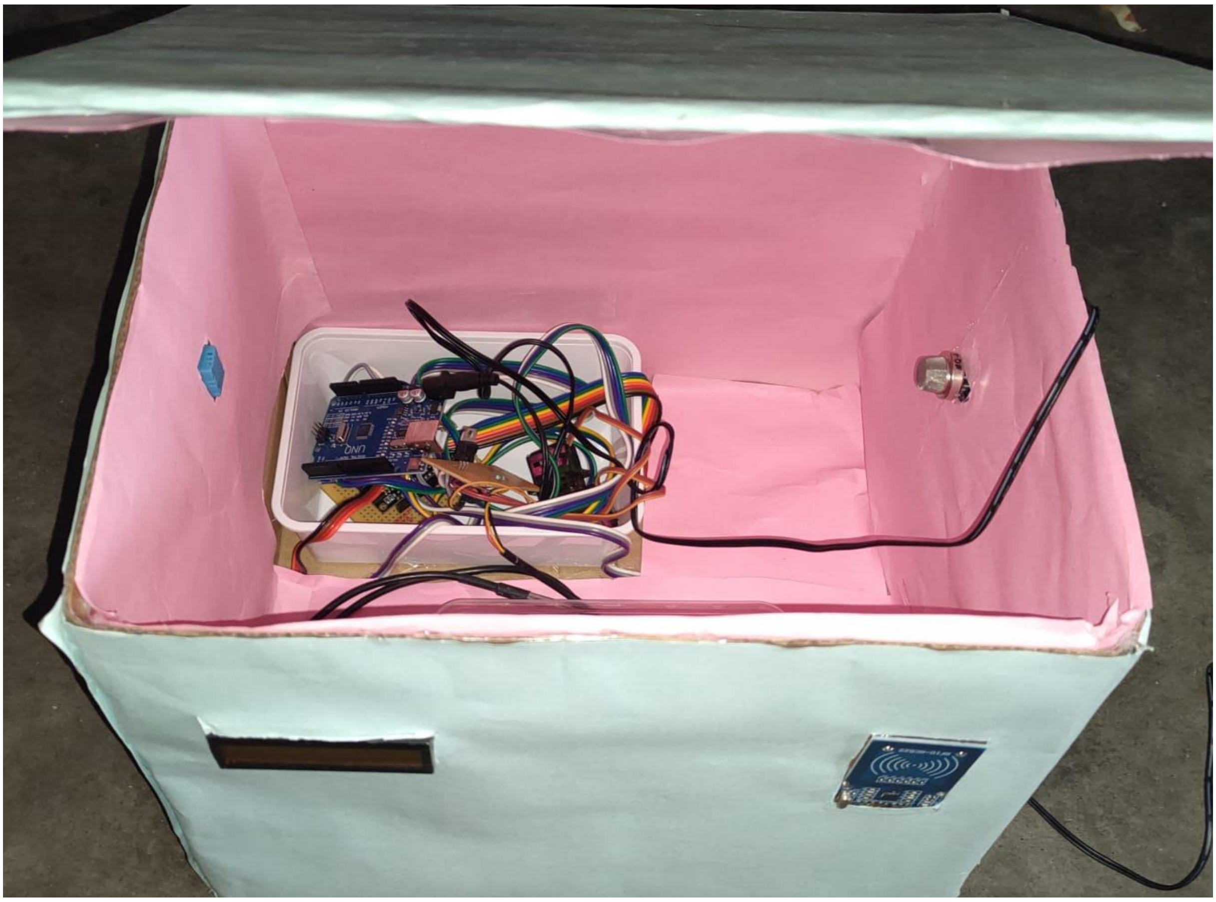
Report: Global Food Losses and Food Waste – Extent, Causes and Prevention

http://www.fao.org/3/i2697e/i2697e.pdf

UN Sustainable Development Goals – Goal 2: Zero Hunger & Goal 12: Responsible Consumption

https://sdgs.un.org/goals

HARDWARE OUTPUT



PROJECT BASED LEARNING

ACADEMIC COURSE **BE23EE408–MEASUREMENTSAND INSTRUMENTATION**

Unit1:Measurement System– NIL

No direct content from this unit was applied in the project.

Unit2:Instruments–NIL

The specific instruments covered in this unit were not directly applied in our setup.

Unit3:Sensors and Transducers– NIL

While this unit was not directly covered, our project employed sensor-transducer combinations such as the DHT11 (temperature and humidity sensor) and MQ-series gas sensors, which internally perform analog-to-digital conversion for microcontroller interfacing. These components indirectly relate to the concepts of sensing and transduction.

Unit 4: Measurement System and Its Applications – Environmental Measurements

This unit directly supported our project. The Smart Food Safety Box relied on real-time measurement of temperature and humidity using the DHT11 sensor, which internally processes analog data and outputs digital values readable by a microcontroller. These sensors helped ensure the internal environment stayed within food safety thresholds. The environmental measurement system was critical in generating alerts for unsafe conditions, demonstrating practical application of measurement concepts covered in this unit.

Unit5:Biomedical Applications– NIL

Not applicable to our current project scope.

ACADEMIC COURSE **BE23EE410–MICROCONTROLLERAND INTERFACING**

Unit1:Introduction to Microcontrollers– NIL

This unit's theoretical aspects were not directly utilized.

Unit2:8051 Microcontroller and Programming –NIL

Nodirectreferencetothe8051microcontrollerwasmadeintheproject.

Unit3:PIC Microcontroller and Programming–NIL

Although we did not use a PIC microcontroller, the general programming principles from this unit helped in writing modular and scalable code for the Arduino UNO, which managed data acquisition and control operations in our project.

Unit4:Communication Interfaces–Sensor and I/O Interfaces

This unit played a central role in our project. We interfaced DHT11 sensors (for temperature and humidity) and MQ gas sensors with the Arduino, which required reading analog signals and converting them using the built-in ADC (Analog-to-Digital Converter) of the microcontroller. The digital data was used to trigger output devices such as buzzers or LEDs, alerting users about unsafe food storage conditions. This integration of input-output and sensor interfacing reflects the core learning outcomes of this unit.

Unit5:RTOS and Real-Time Applications – NIL

Real-time system management using an RTOS was not implemented , but basic task sequencing was handled through Arduino code.